Undemocratic Reversal: Crisis, Recovery, and Weakly Constrained Executive Power in a Laboratory Public-Goods Game

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Abstract

Why do citizens authorize weakly constrained power, and do they reclaim decision rights when its crisis justification recedes? This project tests a behavioral microfoundation of democratic backsliding in a one-session public-goods laboratory experiment in Finland with 400 participants in 40 ten-person pools. Across twenty rounds, anonymously rematched five-person groups vote between citizen-controlled provision and one-person rule. Citizen control preserves individual decision rights but is less productive during the crisis. Leader control raises productivity, compels equal allocations, and permits one participant to move shared points to a personal account. After Block 1, pools are randomized to recovery, which equalizes productivity, or persistence, which preserves the leader's advantage. The standard monetary benchmark favors leader control in both environments, so sustained return to citizen control cannot be explained by that benchmark alone. Repeated ballots, allocations, transfers, beliefs, and payoffs reveal when citizens value retained collective decision rights enough to withdraw their earlier authorization.

See prototype here: `https://panel-lab-experiment.onrender.com`

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1 Research Question and Contribution

Democratic backsliding is usually an elite-led process in which elected incumbents gradually weaken constraints and concentrate authority (Bermeo 2016). Its progress nevertheless depends on a demand side: citizens can punish executive aggrandizement or accept it when partisan, policy, or material considerations outweigh democratic safeguards (Graham and Svolik 2020; Saikkonen and Christensen 2023; Wunsch, Jacob, and Derksen 2025). This proposal studies one behavioral microfoundation of that process. When weakly constrained authority offers a practical response to crisis, will citizens authorize it, and will they withdraw that authorization when its productivity justification recedes?

I call the latter transition *undemocratic reversal*. The term does not describe an entire country reversing a completed episode of backsliding. It identifies a within-person behavioral change: a participant first endorses a rule that concentrates authority and weakens ex ante control, then returns to a rule that preserves citizen decision rights. This focuses the experiment on citizens' role in enabling and resisting executive aggrandizement.

The design contributes in three ways. First, it turns executive constraint into payoff-relevant rules governing interaction. Participants do not merely rate political descriptions. Their ballots determine who controls endowments, how a public good is produced, and whether one person may take a private transfer. The public-good problem supplies a genuine material reason to centralize authority, linking research on democratic restraint to experimental work on delegation and endogenous institutions (Hamman, Weber, and Woon 2011; Kosfeld, Okada, and Riedl 2009).

Second, it identifies a structural source of reversal. All participants enter Block 1 under the same public-service crisis: "**Imagine that you are a citizen in a society where public services have been under serious strain for a long time. Parliament has failed to pass a budget for eight months, healthcare waiting times have doubled, and planning for public services has stalled**." After Block 1, matched pools are randomly assigned to recovery or persistence. Recovery raises the citizen-controlled fund's multiplier to the leader-led rate; persistence retains the crisis rates. The leader's authority, coordination technology, and personal-transfer rule remain fixed.

Third, the design tests whether decision rights carry value beyond the standard monetary benchmark. During the crisis, one-person rule combines greater productivity with compulsory coordination. Recovery removes only the productivity premium: a leader can still overcome voluntary free-riding by requiring equal allocations. Under selfish money maximization, that remaining advantage continues to favor leader control. A return to citizen control therefore runs against the standard benchmark and may reveal an intrinsic value placed on retaining decision rights (Fehr, Herz, and Wilkening 2013; Bartling, Fehr, and Herz 2014). Expectations about payoffs and leader transfers distinguish that value from a belief that voluntary cooperation will now be more profitable.

In simple terms

This proposal asks whether citizens authorize weakly constrained one-person rule when it solves a costly public-service problem, and whether they later take their decision rights back. Recovery raises the citizen-controlled return from 1.50 to 2.50, but a leader can still require equal allocations and prevent free-riding.
Standard money-maximizing behavior therefore favors leader control in both conditions. A return to citizen control after recovery can reveal that decision rights matter beyond earnings. Expected payoffs and transfers help determine whether citizens reclaim control because they value those rights or because they expect citizen cooperation to pay more.

2 Theory

2.1 Conditional tolerance of weakly constrained authority

Citizens may accept weak executive constraints without rejecting democracy in principle. When existing institutions fail to deliver valued outcomes, material and policy considerations can outweigh safeguards on executive power (Kingzette et al. 2021; Saikkonen and Christensen 2023; Wunsch, Jacob, and Derksen 2025). Crisis changes the practical price of retaining decision rights.

The citizen-led procedure (D) preserves individual control: each person chooses how many points enter a public-services fund, and none can be removed for personal use. In Block 1, however, the fund is relatively unproductive. Under the leader-led procedure (A), one person decides for all five citizens, the fund is more productive, and the leader may move points to a personal account. A vote for A can thus be an instrumental response to crisis.

H1: Crisis authorization. In Block 1, support for the leader-led method will be greater among participants who expect its productivity and coordination advantages to outweigh the value of retained decision rights and the risk of a personal transfer.

2.2 A behavioral microfoundation of backsliding

Backsliding is not reproduced wholesale in a five-person laboratory game. The experiment isolates a prior behavioral step: citizens authorize weaker constraints, after which a leader decides how to use the resulting discretion. A ballot for A lets one participant decide for everyone and accepts the possibility of a personal transfer; the transfer ρ records how the selected leader uses that authority.

The transfer is neither described nor interpreted as literal corruption. Its purpose is to make weak constraints consequential: authority can improve provision, but it also gives another participant control over the distribution of shared resources. Later ballots show whether performance and observed use of discretion alter support for that authority in a transparent environment suited to the Finnish context.

H2: Use of discretion. Some selected leaders will move public-service points to their personal accounts, and larger observed transfers will be followed by lower support for the leader-led method among exposed participants.

2.3 Structural change and democratic reversal

Recovery changes the institutional bargain. The citizen-led method becomes as productive as the leader-led method, removing the latter's exogenous productivity premium while preserving its compulsory coordination and discretion over transfers. Citizen control still risks free-riding; leader control still limits individual choice. Returning to D can therefore reveal how much participants value decision rights once the crisis no longer makes those rights comparatively costly.

Let $\lambda_i \geq 0$ denote participant i's nonmonetary value of retained decision rights and let $\gamma_i \geq 0$ capture aversion to the leader's expected transfer. Participant i prefers D when

$$\mathbb{E}_i[\pi^D - \pi^A] + \lambda_i + \gamma_i \mathbb{E}_i[\rho] > 0. \tag{1}$$

Recovery changes the expected-payoff term; decision-right value and the leader's formal authority remain fixed. The standard monetary benchmark sets $\lambda_i = \gamma_i = 0$, whereas behavioral reversal can arise when the changed material price brings the nonmonetary value of control into the institutional choice.

Persistence supplies the counterfactual: participants complete the same session and rounds, but citizen-led productivity does not improve. The contrast separates recovery from experience, fatigue, end-game behavior, and regression to the mean, without return-visit attrition.

H3: Payoff updating. Recovery will raise the expected payoff from citizen control relative to leader control and reduce the perceived need for one-person authority more than persistence.

H4: Democratic reversal. Among participants who choose A in Block 1's final round, assignment to recovery will increase the probability of choosing D in Block 2's final round despite the leader-led method's remaining coordination advantage.

H5: Value of decision rights. Among participants assigned to recovery, some reversals will occur even when A is expected to pay at least as much as D. Such choices provide the sharpest behavioral evidence that retained decision rights enter utility beyond their monetary return.

3 Experimental Design

3.1 Overview

In one synchronized 75–90 minute session, participants complete two ten-round blocks within stable ten-person matching pools. Before each paid round, pools are anonymously reshuffled into two five-person groups. After Block 1, pools are randomized to a Block-2 condition and shown the applicable fund rates, allowing repeated interaction without reliable bilateral reputations.

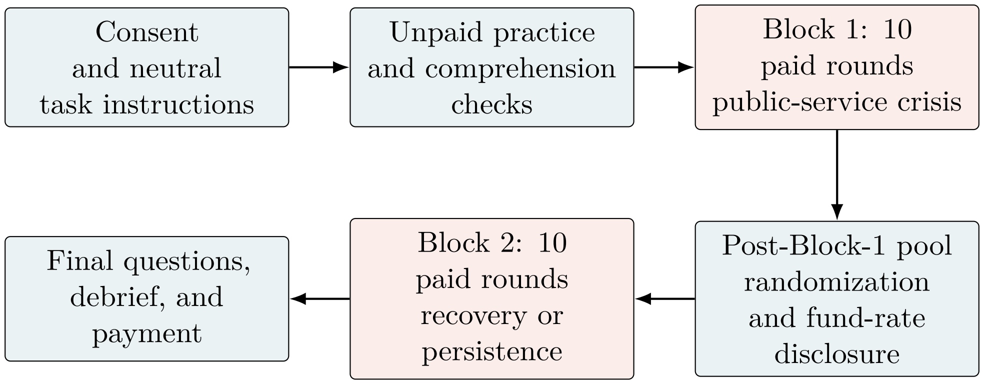

Figure 1: One-session sequence. Each paid round includes an individual choice, allocation, feedback, and anonymous rematching.

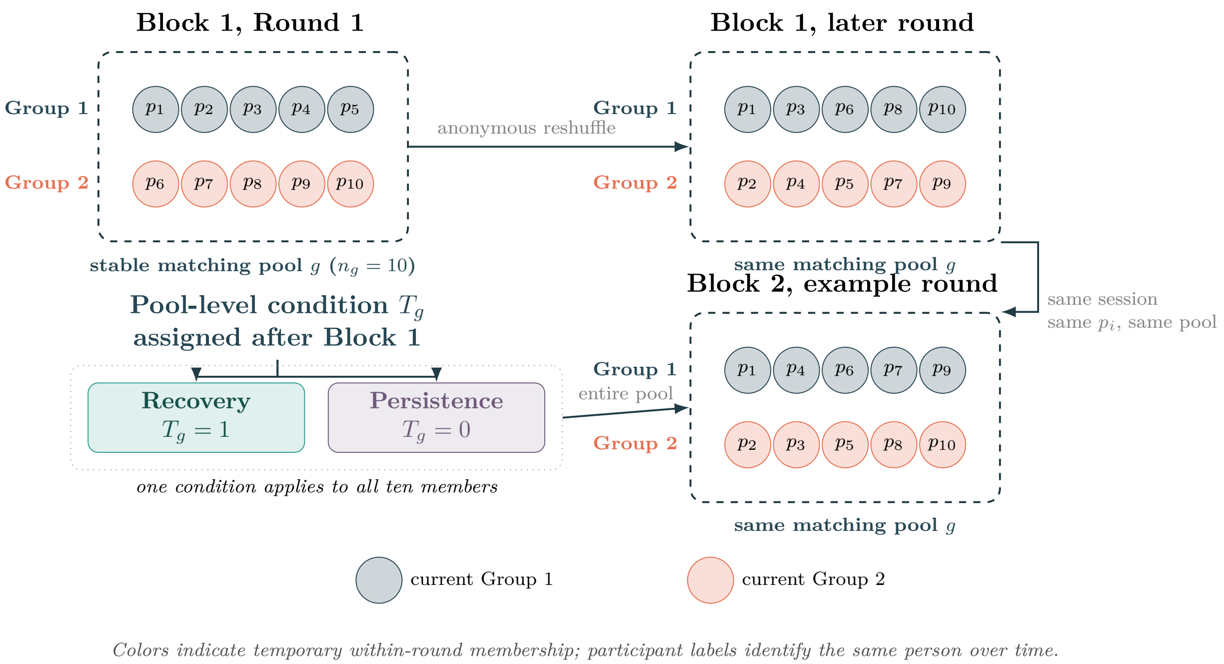

Figure 2: Matching pools, anonymous rematching, and Block-2 treatment assignment.

In simple terms

Participants remain in a ten-person pool (g) but are reshuffled into two temporary five-person groups before each round. They never play outside their pool. In Figure 2, colors show current group membership, whereas p_i follows the same person over time. After Block 1, the whole pool receives recovery ($T_g = 1$) or persistence ($T_g = 0$); the design includes 40 pools.
This creates interaction while limiting stable reputations, collusion, and retaliation. It also contains spillovers within one treatment and guarantees two complete groups per round. Repeated exposure within pools remains the cost.
Reversal (R_i) is measured for participant p_i, but treatment (T_g) is assigned to pool g. Inference reproduces that pool-level assignment, with pool-clustered estimates as robustness checks;

temporary groups organize play but are not treatment units.

3.2 The institutional-choice public-good game

Each round begins with 20 points for each of five group members. Every member privately votes between the citizen-led procedure (D) and the leader-led procedure (A). A simple majority of at least three votes selects the binding method. Individual ballots remain secret; the group sees the aggregate vote and the selected rule. The neutral labels shown to participants are "Each person chooses" and "One person chooses for the group."

Citizen-led procedure (D). Each member chooses an allocation c_i of 0–20 points to a public-services fund; unallocated points remain in the member's personal account. Let m_{gb} denote the fund's productivity for matching pool g in block b. Player i earns

$$\pi^D_{igbr} = 20 - c_{igbr} + \frac{m_{gb}}{5} \sum_{j=1}^{5} c_{jgbr}. \tag{2}$$

The marginal private return remains below 1 and the group return above 1. Here, $m_{g1} = 1.50$; in Block 2, $m_{g2} = 2.50$ under recovery and 1.50 under persistence.

Leader-led procedure (A). The leader is chosen among members who have served least often, with ties broken randomly. That person sets a common allocation $t \in \{0, \ldots, 20\}$ and may move $\rho \in \{0, \ldots, 20\}$ fund points to a personal account, subject to $\rho \leq 5t$. The remaining fund $5t - \rho$ always uses multiplier 2.50. A nonleader earns

$$\pi^A_{igbr} = 20 - t + \frac{2.50}{5}(5t - \rho), \tag{3}$$

while the leader earns

$$\pi^{A,L}_{igbr} = 20 - t + \frac{2.50}{5}(5t - \rho) + \rho. \tag{4}$$

Members observe the required allocation, transfer, public-service return, and own payoff. Leader control coordinates provision and, in Block 1, is more productive, but places distribution in another participant's hands.

How incentives map onto reversal. In the crisis, the citizen-led fund uses $m = 1.50$ and the leader-led fund $m = 2.50$. Recovery raises the former to 2.50 while leaving the leader rule unchanged; persistence retains the crisis rates and identifies changes from repeated play, fatigue, or end-game behavior. The relevant benchmark is not a fixed illustrative allocation but the behavior implied by standard money maximization.

Table 1: Monetary benchmark and full-provision outcomes

Outcome	D: crisis	D: recovery	A: either block
Selfish monetary prediction	$c_i = 0$; each earns 20.	$c_i = 0$; each earns 20.	$t = 20, \rho = 20$; non-leaders earn 40, the leader earns 60, and the ex ante mean is 44.
Full provision without transfer	$c_i = 20$; each earns 30.	$c_i = 20$; each earns 50.	$t = 20, \rho = 0$; each earns 50.

Note: The ex ante mean assumes each citizen has a one-fifth chance of serving as leader. Persistence retains the crisis column for D in Block 2.

Under D, a contributed point returns 0.30 to its contributor in the crisis and 0.50 after recovery, so $c_i = 0$ is privately optimal in both states. Under A, the leader's payoff is $20 + 1.5t + 0.5\rho$, making $t = 20$ and $\rho = 20$ privately optimal. The resulting monetary benchmark ranks leader control above citizen control in both blocks. Recovery improves the cooperative frontier of D but does not alter its free-riding equilibrium.

This distinction makes reversal theoretically informative. Conditional cooperation can make recovered D profitable, while an intrinsic value of decision rights, aversion to unequal transfers, or both can make it desirable beyond earnings (Fehr, Herz, and Wilkening 2013; Bartling, Fehr, and Herz 2014). Before the final Block-1 ballot and the first and final Block-2 ballots, participants report expected payoffs under both methods and the expected leader transfer. These measures identify whether a reversal accompanies a changed monetary ranking or occurs despite an expected payoff advantage for A.

In simple terms

Each round, five citizens receive 20 points and cast private ballots ($V_{ibr} \in \{\mathsf{D}, \mathsf{A}\}$). A majority selects the method. Under "Each person chooses" (D), each citizen chooses a fund allocation (c_{igbr}); the total is multiplied by the current rate (m_{gb}), divided equally, and unallocated points remain personal.
Under "One person chooses for the group" (A), the software selects among those who have served least often. Citizens choose the method, not a candidate. The selected decision-maker sets everyone's allocation (t) and may move fund points to their own payoff ($\rho \leq \min\{20, 5t\}$). The remainder ($5t - \rho$) is multiplied by 2.50 and shared equally; the decision-maker keeps ρ. Everyone then sees the decision, fund, transfer, return, and own payoff (π^D_{igbr} or π^A_{igbr}). The group dissolves, and participants are rematched within pool g.

Table 2: Implemented design parameters

Feature	Implemented value
Laboratory session	One synchronized visit, approximately 75–90 minutes
Paid rounds	10 per block (20 total), preceded by unpaid practice
Group and matching pool	5-person groups; 10-person treatment-matched pools
Rematching	Anonymous rematching within pool before every paid round
Round endowment	20 points per participant
Citizen-led multiplier	1.50 in Block 1; 2.50 recovery / 1.50 persistence in Block 2
Leader-led multiplier	2.50 in both blocks and conditions
Leader's personal transfer	0–20 fund points, limited by the number of points in the fund
Performance payment	One randomly selected game round from each block
Exchange rate	EUR 0.20 per point
Show-up payment	EUR 10.00

The random-round rule limits wealth effects, hedging across rounds, and excessive payment while preserving incentives in every decision.

3.3 Block 1: crisis and leader authorization

After consent, participants receive short, neutral on-screen instructions and active unpaid practice on both methods and their payoff rules. The interface labels the methods "Each person chooses" and "One person chooses for the group" and does not identify either method with democracy or backsliding. Three comprehension questions precede paid play; development configurations allow omissions. Block 1 presents a prolonged public-service crisis: each fund point produces 1.50 under D and 2.50 under A, whose selected decision-maker may move up to 20 fund points to their own payoff.

Ten paid rounds follow: individual choice, majority selection, method-specific fund decision, feedback, and rematching. Expected payoffs and transfers are elicited before the final choice. No democratic-constraint questions are asked before or between the paid blocks. The software records choices, allocations, transfers, and payoffs, and randomly selects one round for payment.

3.4 Block 2: structural change and reversal

After every participant completes Block 1, pools are randomized to *recovery*, which raises the citizen-led multiplier from 1.50 to 2.50, or *persistence*, which leaves it at 1.50. The leader-led multiplier, authority, and maximum transfer remain fixed. The transition page reports the applicable numerical rates without adding an evaluative recovery vignette.

Participants report expected payoffs and transfers before the first choice, complete ten otherwise identical rounds, and report those expectations again before the final choice. The first choice measures immediate revision; the last measures sustained reversal and is the primary endpoint. Only after that endpoint is recorded do participants complete the public-service and democratic-constraint battery. One Block-2 round is randomly paid, and both selected payoffs are converted to euros and added to the participation payment.

4 Outcomes, Process Measures, and Estimands

Let $V_{ibr} \in \{\mathsf{D}, \mathsf{A}\}$ be participant i's ballot in block b and round r. Define Block-1 leader endorsement as

$$A_{i1} = \mathbb{1}\{V_{i,1,10} = \mathsf{A}\}. \tag{5}$$

For the theoretically relevant subgroup with $A_{i1} = 1$, democratic reversal is

$$R_i = \mathbb{1}\{V_{i,2,10} = \mathsf{D}\}. \tag{6}$$

Let $T_g = 1$ indicate that the public-service crisis receded for matching pool g. The primary causal estimand is

$$\tau_R = \mathbb{E}[R_i(1) - R_i(0) \mid A_{i1} = 1], \tag{7}$$

an intention-to-treat contrast. Because treatment is randomized only after Block 1, A_{i1} is unambiguously pretreatment. The primary test uses randomization inference that reproduces the pool-level assignment; an unadjusted difference in means and regression estimates with pool-clustered and wild-cluster uncertainty will also be reported.

In simple terms

The primary estimand (τ_R) asks: among participants who choose "One person chooses for the group" in Block 1's final round ($V_{i,1,10} = \mathsf{A}$; $A_{i1} = 1$), how much does recovery ($T_g = 1$), rather than persistence ($T_g = 0$), increase a final-round return to "Each person chooses" ($V_{i,2,10} = \mathsf{D}$; $R_i = 1$)? Under recovery, both methods are equally productive, but one decision-maker can still require equal allocations, so reversal accepts continued free-riding risk.
We cannot observe the same person's choice under both recovery ($R_i(1)$) and persistence ($R_i(0)$). Post-Block-1 random assignment therefore identifies τ_R by comparing reversal rates across assigned conditions (T_g). The assignment and statistical test both operate at pool level.

The ballot measures authorization; the transfer ρ_{igbr} measures its use. Among selected leaders, a personal transfer is $M_{igbr} = \mathbb{1}\{\rho_{igbr} > 0\}$, with intensity recorded in points and as $\rho_{igbr}/(5t_{igbr})$ when $t_{igbr} > 0$. Because leader selection and prior outcomes are endogenous, these are behavioral process measures, not randomized recovery effects.

Immediate reversal is

$$R_i^I = \mathbb{1}\{V_{i,2,1} = \mathsf{D}\} \quad \text{for } A_{i1} = 1. \tag{8}$$

It records the first response to the new rates, before Block-2 interaction. The primary R_i records whether that response is sustained after ten rounds. Other secondary outcomes are the change in leader-vote share across Rounds 8–10, the round-by-treatment trajectory, and group-level method choice.

Expected payoffs under both methods and the expected transfer are measured before the final Block-1 choice and the first and final Block-2 choices. These expectations distinguish reversals accompanied by an expected monetary gain from those that preserve an expected payoff advantage for A. Public-service beliefs, perceived risk to constraints, and the seven-item index are measured only after the final behavioral outcome, preventing those explicitly political questions from cueing earlier choices. They are descriptive endline measures rather than pretreatment moderators; choices, decisions, and payoffs are recorded by round.

5 Randomization, Sample, and Power

The sample comprises 400 participants in 40 ten-person pools. Treatment is assigned only after every participant has completed Block 1. Within each session, pools are ordered by their final-round Block-1 leader-vote share and paired; one pool in each pair is randomly assigned to recovery and the other to persistence. An unmatched pool is assigned by a fair random draw. This preserves random assignment while balancing the pretreatment prevalence of the subgroup defining reversal. A single visit removes panel attrition, while overbooking and standby participants protect the complete five-person groups required at launch.

Power depends on Block-1 endpoint support for A, the persistence-group reversal rate, and intrapool correlation. Pilot estimates will inform pool-level simulations. Repeated rounds improve measurement and reveal trajectories; they do not create 8,000 independent observations from 400 participants in 40 randomized pools.

Randomization and confirmatory inference both operate on whole pools. The primary randomization test permutes treatment within the pairs created after Block 1 and reproduces the fair draw for unmatched pools. Pool-clustered regression and wild-cluster inference provide robustness checks. Production sessions require multiples of ten; concurrent rooms randomize and match within cohort, preventing interaction across treatments.

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